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Week 2 Progress Report

Website Development & Database Fundamentals

Weekly goal: In this week I'll created a professional portfolio website , installed the development tools, and built own local WordPress environment. This week shifts from planning to building: you will turn your project idea into a real website, and learn how WordPress stores information inside a database.

Introduction

The second week of my internship focused on learning **WordPress website development** and understanding the complete website development workflow. Throughout the week, I explored the WordPress interface, learned how different components work together, and developed a portfolio website by applying the concepts learned during the training sessions.

Learning Objectives

The primary objective of this week was to understand the fundamentals of website design and development using WordPress. The focus was on learning how to create a professional, user-friendly, and responsive website by following modern web design principles and industry standards.

Activities Performed

During this week, I gained hands-on experience with the WordPress environment. I learned how to set up the local development environment using **WAMP Server**, start the Apache and MySQL services, and access the WordPress dashboard through the localhost server.

I explored the complete WordPress interface, including the Dashboard, Pages, Posts, Media Library, Themes, Plugins, Appearance settings, Menus, Widgets, and the Block Editor. I also learned the purpose of each feature and how they are used during website development.

Before designing the website, I studied the structure of a professional homepage and understood the importance of planning a website before implementation. A simple wireframe was prepared to define the layout and organization of different website sections. This planning process helped in creating a structured and organized user interface.

Using the WordPress Block Editor, I designed the homepage of my portfolio website by creating the following sections:

- Navigation Bar
- Hero Banner
- About Section
- Services / Features Section
- Call-to-Action (CTA)

- Footer

Special attention was given to maintaining a clean layout, proper spacing, visual hierarchy, and user-friendly navigation to provide a better user experience.

In addition, I learned how each homepage section serves a specific purpose. The Hero section was designed to capture the visitor's attention, the About section introduced the website and its purpose, the Features section highlighted key services, the Call-to-Action encouraged user interaction, and the Footer contained important information such as contact details and copyright.

Screenshots of both the initial wireframe and the completed homepage were documented for future reference and portfolio purposes.

Day 2: Website Customization and CSS Styling

On the second day of the internship, I learned how to enhance the visual appearance of a website using **CSS (Cascading Style Sheets)** and customize a WordPress theme. I understood that while HTML provides the structure of a webpage, CSS is responsible for its overall design, including colors, typography, spacing, buttons, borders, and layout.

I explored the **WordPress Theme Customizer** and gained hands-on experience with different customization options such as **Site Identity, Colors, Typography, Header, Footer, and Navigation Menus**. I also learned how to apply custom CSS through the **Additional CSS** section to modify the appearance of website components according to design requirements.

During this activity, I customized my portfolio website by selecting a consistent color palette, updating the website logo, modifying typography, designing buttons, and improving the header and footer sections. I also prepared a simple branding guideline by selecting primary and secondary colors along with suitable fonts to maintain a consistent visual identity throughout the website.

Skills Learned

- Website Branding Fundamentals
- CSS Basics
- WordPress Theme Customization
- Typography Selection
- Color Palette Selection
- Button Styling
- Header and Footer Customization
- UI Consistency Principles

Day 3: Understanding Databases and phpMyAdmin

The third day focused on understanding how **WordPress stores website data inside a MySQL database**. I learned that every page, post, user account, and website setting is stored in different database tables instead of static files.

Using **phpMyAdmin**, I explored the WordPress database and became familiar with important tables such as **wp_posts**, **wp_users**, **wp_comments**, **wp_options**, and **wp_terms**. I learned the purpose of each table and how WordPress retrieves information whenever a webpage is opened.

I also executed my first SQL queries to retrieve information from the database. By using different **SELECT** statements, I learned how to display all records, retrieve only specific columns, and count the total number of records stored in a table. This practical exercise helped me understand the relationship between the WordPress dashboard and the underlying database.

Skills Learned

- Database Fundamentals
- MySQL Basics
- phpMyAdmin Interface
- WordPress Database Structure
- SQL SELECT Queries
- Database Exploration

Day 4: SQL Practice Using LocalWP and Adminer

On the fourth day, I continued learning SQL by using **LocalWP** and **Adminer**, which provided a safe environment for exploring the WordPress database without affecting the actual website.

I practiced several SQL queries to retrieve, filter, sort, and summarize data. During this session, I learned the practical use of SQL commands such as **SELECT**, **WHERE**, **ORDER BY**, **COUNT**, **LIMIT**, **JOIN**, and **GROUP BY**.

To reinforce my understanding, I created a temporary WordPress page, verified its existence in the database through SQL queries, and observed how the database updated automatically whenever the page was published or removed. I also learned how **JOIN** combines related data from multiple tables and how **GROUP BY** summarizes records into meaningful categories.

This session greatly improved my understanding of how websites interact with databases and how SQL is used to retrieve business information efficiently.

Skills Learned

- SQL Query Writing
- Data Filtering
- Data Sorting
- SQL JOIN

- SQL GROUP BY
- Database Relationships
- LocalWP
- Adminer

Day 5: Understanding How WordPress Stores Information

The final day of the week focused on understanding how WordPress automatically performs database operations whenever a user interacts with the website.

I learned that publishing a new page performs an **INSERT** operation, updating an existing page performs an **UPDATE** operation, and deleting a page performs a **DELETE** operation within the database. Although these SQL commands are executed automatically by WordPress, understanding their functionality helped me visualize what happens behind the scenes whenever content is managed through the dashboard.

I practically created a new page, modified its title, and then moved it to the trash while observing how each action was reflected in the database. This activity helped me connect the concepts of SQL with real-world WordPress operations.

At the end of the week, I reviewed all the topics covered, including website development, theme customization, database exploration, and SQL fundamentals. I also updated my GitHub repository with the week's progress and documented the work completed during the second week of the internship.

Skills Learned

- SQL INSERT Concept
- SQL UPDATE Concept
- SQL DELETE Concept
- WordPress Data Management
- Database Synchronization
- CRUD Operations
- GitHub Documentation
- Weekly Progress Reporting

Skills Learned

During the second week of my internship, I developed the following technical skills:

- Understanding the WordPress Content Management System (CMS)
- Setting up a local development environment using WAMP Server and LocalWP
- Designing responsive web pages using the WordPress Block Editor

- Creating website wireframes before implementation
- Developing a professional portfolio website
- Customizing WordPress themes
- Applying CSS for website styling and branding
- Creating consistent color palettes and typography
- Designing website navigation, hero section, footer, and call-to-action sections
- Understanding the structure of MySQL databases
- Exploring WordPress databases using phpMyAdmin and Adminer
- Understanding the purpose of WordPress database tables
- Writing and executing basic SQL queries
- Using SQL commands such as **SELECT, WHERE, ORDER BY, COUNT, LIMIT, JOIN, and GROUP BY**
- Understanding CRUD (Create, Read, Update, Delete) operations in WordPress
- Understanding how WordPress automatically stores and updates data in the database
- Maintaining project documentation and updating progress through GitHub

Conclusion

The second week of my internship focused on developing both frontend and database fundamentals. I learned how to design and customize professional WordPress websites using themes and CSS while also gaining practical knowledge of MySQL databases and SQL queries. Through hands-on activities with phpMyAdmin, LocalWP, and Adminer, I developed a clear understanding of how WordPress stores, retrieves, and manages data behind the scenes. These concepts provided a strong technical foundation that will support more advanced web development and backend programming in the upcoming weeks.



